

Analog and Digital Communication Lab

Laboratory report submitted for the partial fulfillment
of the requirements for the degree of

Bachelor of Technology
in
Electronics and Communication Engineering

by

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Experiment No. - 08

0.1 Aim

1. Hardware implementation of BPSK modulation and its non-coherent demodulation.

0.2 Apparatus Used

1. ICs: XR-2206 (Monolithic F.G.)
2. Diode (1N4007)
3. DC Power Supply
4. Breadboard
5. LM741 (Op - Amp)
6. Resistance
7. Connecting Wires
8. Function Generator
9. LM393 (Voltage Comparator)
10. Capacitor
11. DSO Probes

0.3 Theory

0.3.1 BFSK Modulation (Binary Frequency Shift Keying)

Binary Frequency Shift Keying (BFSK) refers to a form of frequency modulation that utilizes two unique frequencies to convey binary information (0s and 1s). Each bit within the data stream is assigned to one of the two frequencies, with one frequency representing logic 1 and the alternate representing logic 0. Basic Concept:

- In BFSK, two different frequencies f_1 and f_2 are used to transmit binary data.
- A binary 0 is represented by frequency f_1 , and a binary 1 is represented by frequency f_2 .

Mathematical Representation:

$$s(t) = \begin{cases} A \cos(2\pi f_1 t), & \text{for bit '0'} \\ A \cos(2\pi f_2 t), & \text{for bit '1'} \end{cases} \quad (1)$$

where:

- A is the amplitude of the signal.
- f_1 and f_2 are the frequencies used to represent binary 0 and 1, respectively.
- The signal switches between f_1 and f_2 according to the bits of the data stream.

Frequency Separation:

For BFSK, the two frequencies, f_1 and f_2 should be sufficiently separated to ensure clear differentiation at the receiver, often by a factor known as the "frequency separation" Δf . This is given by:

$$\Delta f = |f_2 - f_1| \quad (2)$$

A larger Δf helps in reducing errors, especially in noisy environments.

Bandwidth Considerations:

The bandwidth of the BFSK signal depends on the frequency separation Δf , base band signal frequency f_m , and the modulation index. The bandwidth can be approximated as:

$$BW_{BFSK} \approx \Delta f + 2f_m \quad (3)$$

0.3.2 Non-Coherent Demodulation of BFSK

Non-coherent demodulation describes a technique in which the receiver does not require any prior knowledge of the carrier phase, meaning there is no synchronization regarding the carrier's phase between the transmitter and receiver. This approach is advantageous in real-world systems where achieving phase synchronization can be challenging or expensive.

0.3.3 FSK Modulator Circuit and Block Diagram of FSK Demodulation

The FSK modulator circuit is used to generate Frequency Shift Keying signals by switching between two frequencies corresponding to binary inputs. The XR2206 IC is commonly used for generating FSK signals by varying frequency according to the input message signal.

The block diagram of FSK demodulation consists of band-pass filters, envelope detectors, and a voltage comparator. The incoming FSK signal is passed through separate band-pass filters tuned to the two frequencies, followed by envelope detection and comparison to recover the original binary data.

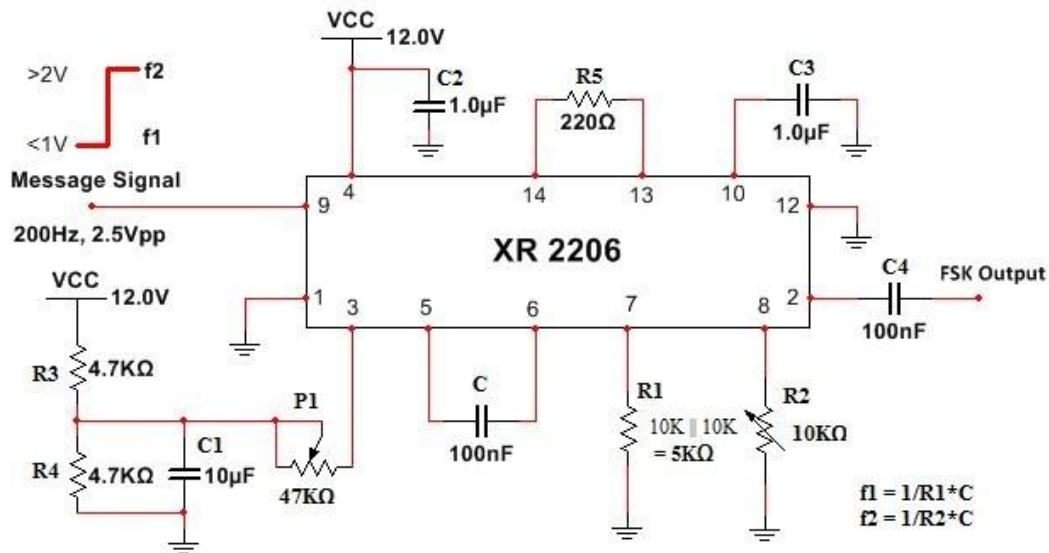


Figure 1: FSK Modulator Circuit

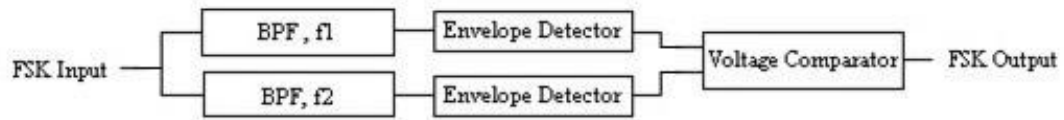


Figure 2: Block Diagram of FSK Demodulation

0.3.4 Band Pass Filter Circuit and Designing Equations

$C_1 = C_2 = C = 0.01\mu F; BW_{BPF} = 500Hz$					
1.	$R_1 = \frac{Q}{2\pi f_c C A_f}$	3.	$R_3 = \frac{Q}{\pi f_c C}$	5.	$A_f < 2Q^2$
2.	$R_2 = \frac{Q}{2\pi f_c C (2Q^2 - A_f)}$	4.	$Q = \frac{f_c}{BW}$	6.	$A_f = \frac{R_3}{2R_1}$

Table 1: BPF Designing Equations

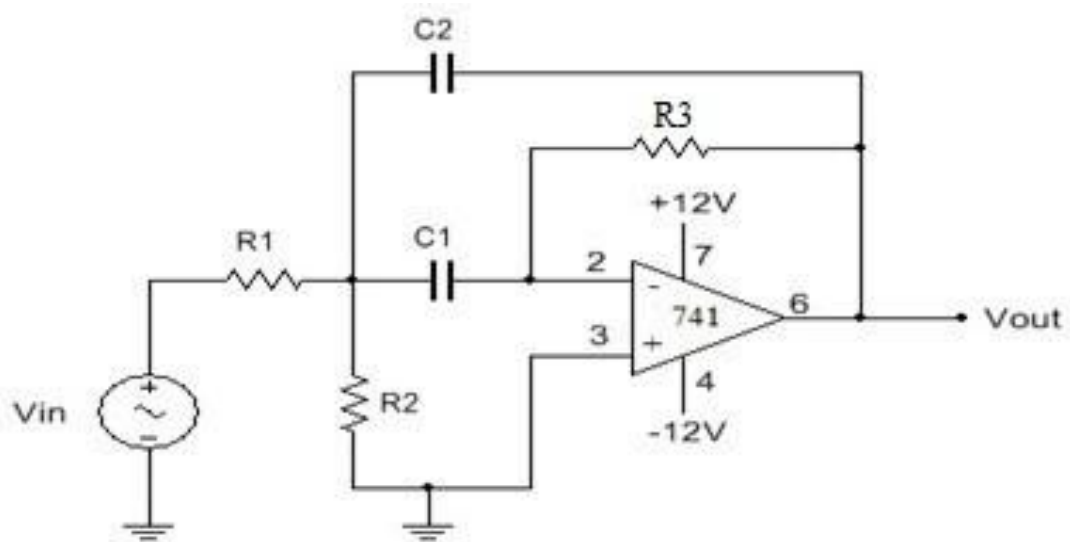


Figure 3: Band Pass Filter

0.3.5 Envelope Detector Circuit

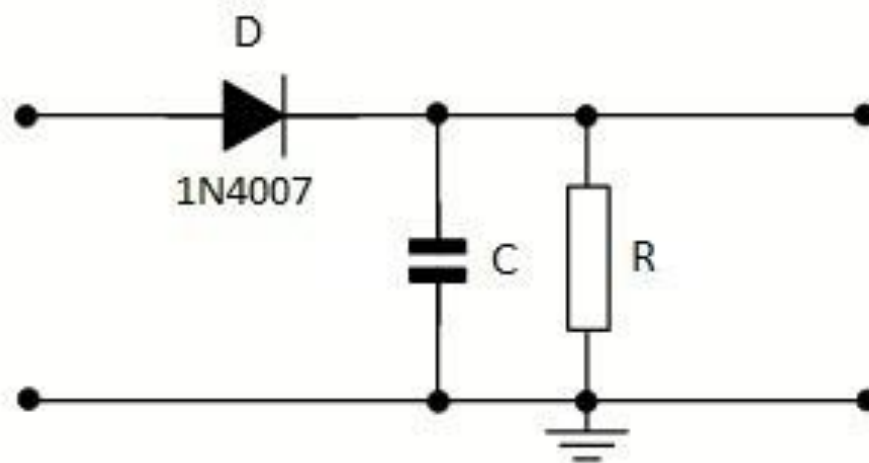


Figure 4: Envelope Detector Circuit

0.4 Procedure

(FSK Modulation)

1. Read IC data sheet (XR-2206), and then choose suitable circuit for FSK modulation.
2. Connect the FSK Modulator circuit shown in Figure 1 and give square wave signal of 200 Hz and 2.5 Vpp as input.
3. As the formula given in the data sheet calculate R1 and R2 with the given carrier signals frequencies, $f_1 = 2$ KHz and $f_2 = 5.2$ KHz.
4. Observe the time and frequency domain wave forms of FSK modulator output. Draw the output and analyze it in both domains.

(FSK Demodulation)

5. Design the Band pass filter for different carrier frequencies and envelope detector circuit. Connect as shown in Figure 2.
6. Give a FSK Modulated signal as input.
7. Observe FSK Demodulated output and draw it in your copy, compare it with message signal.

0.5 Observations

0.5.1 Output Images

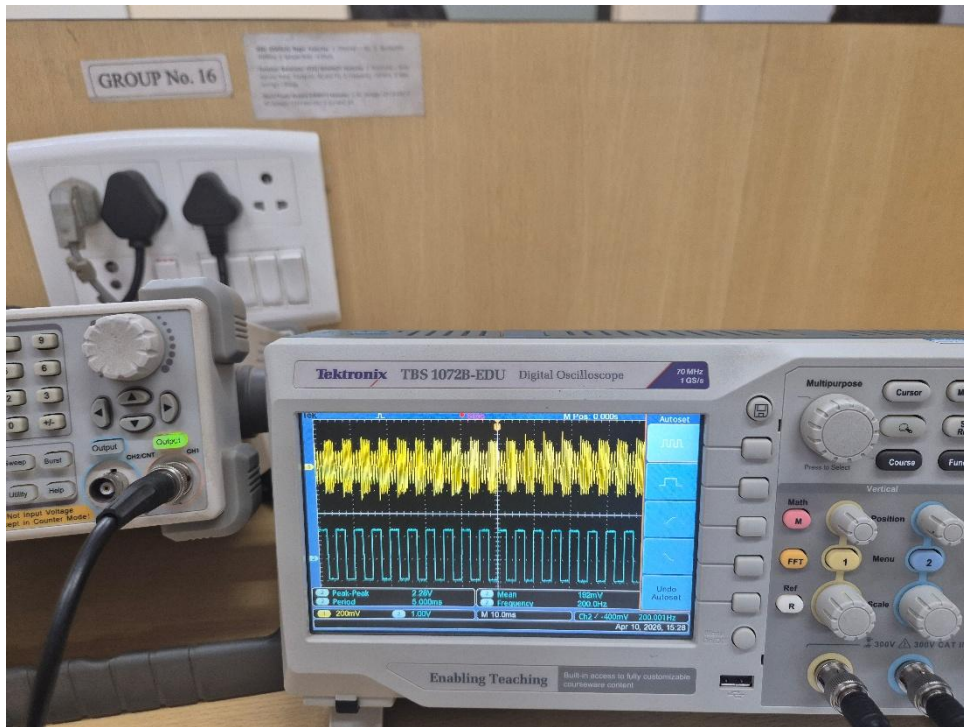


Figure 5: Time Domain of Modulating Signal & BFSK Modulated Signal

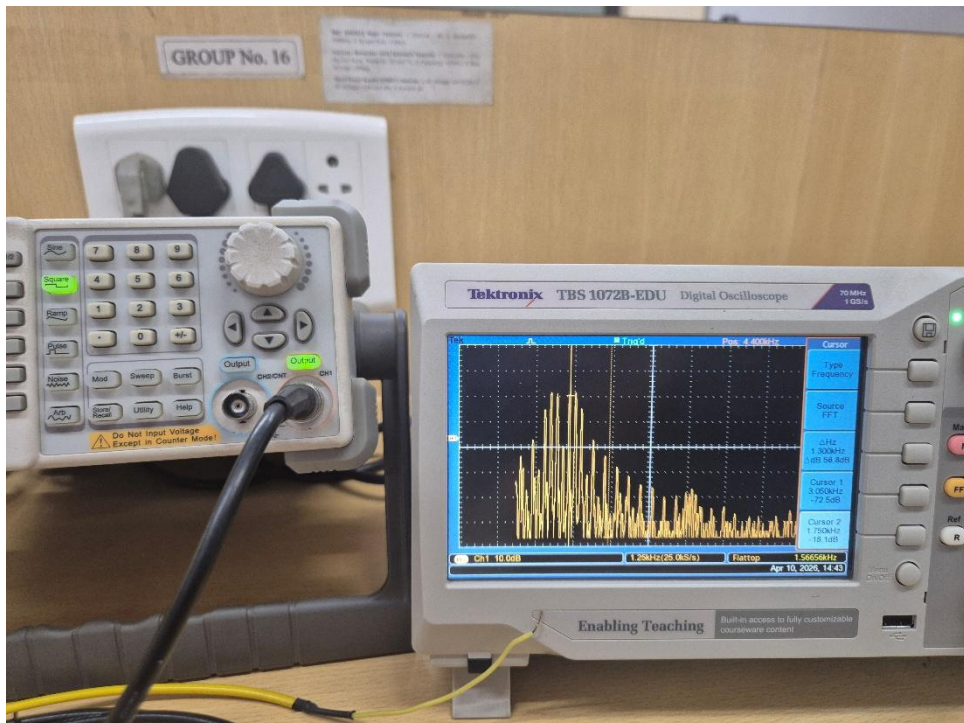


Figure 6: Frequency Domain of BFSK Modulated Signal at Frequency f_1 , f_2
 $f_1 = 2\text{GHz}$
 $f_2 = 5.2\text{GHz}$



Figure 7: BPF Output at Frequency 5.2GHz

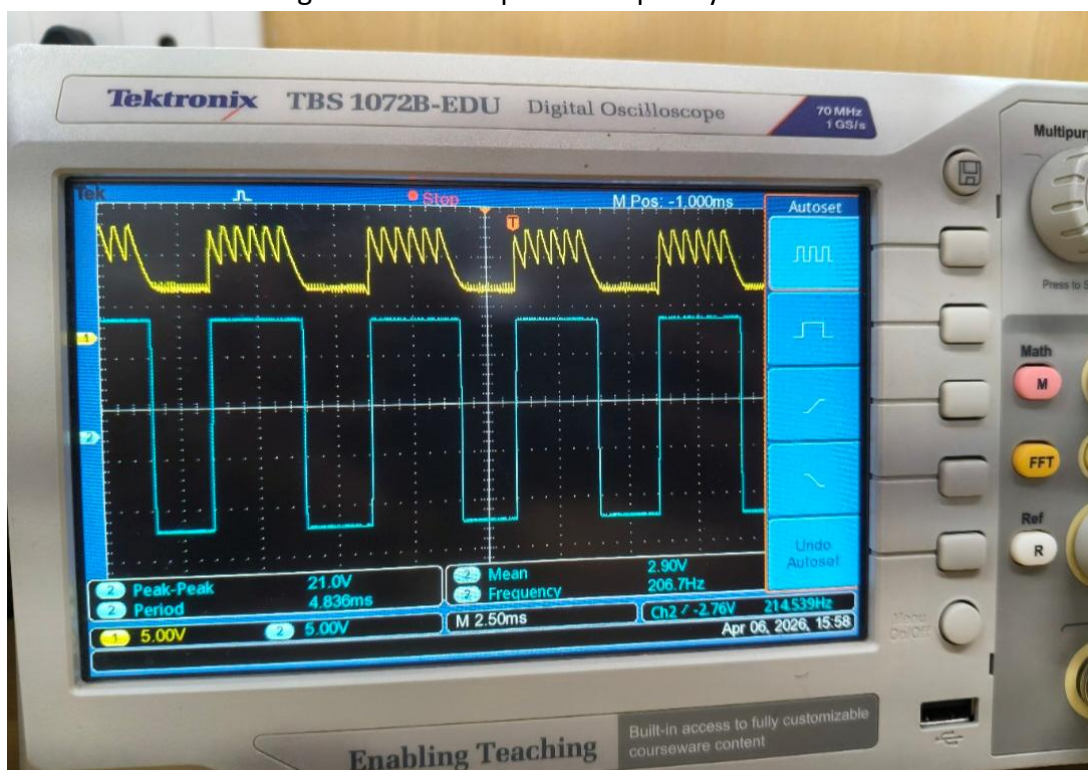


Figure 8: Time Domain of Envelope Detector Output

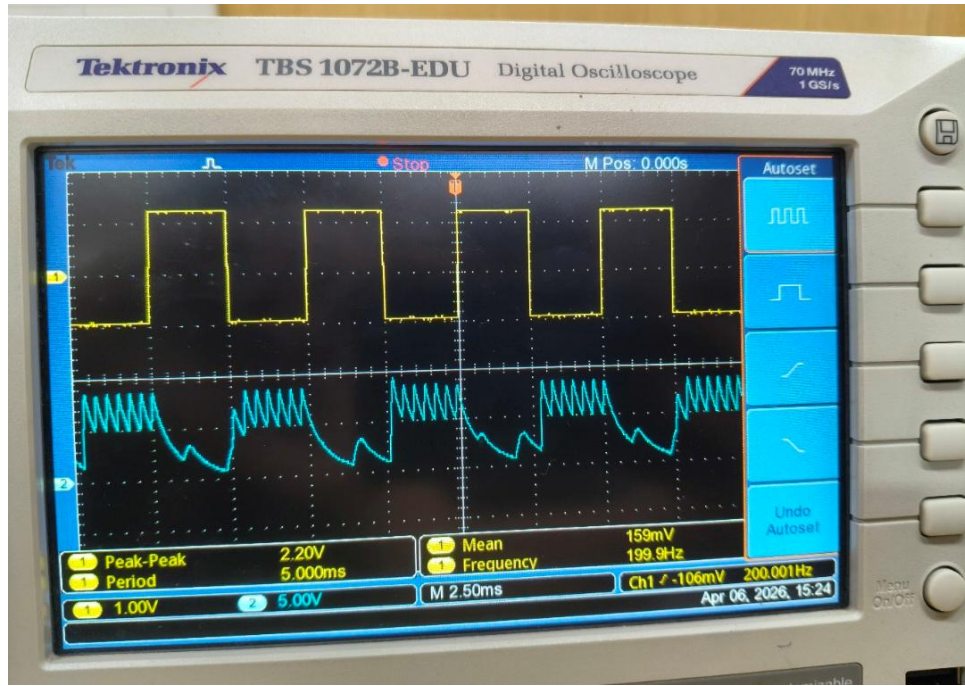


Figure 9: Time Domain of Voltage Comparator Output

0.5.2 Calculation of Band Pass Filter Parameters

The design calculations for the Band Pass Filter are performed using the standard BPF design equations.

Given:

$$f_c = 5200 \text{ Hz}, \quad BW = 500 \text{ Hz}, \quad C = 0.01 \mu\text{F}$$

Quality Factor (Q):

$$Q = \frac{f_c}{BW} \quad (4)$$

$$Q = \frac{5200}{500} = 10.4 \quad (5)$$

Gain Factor (A_f):

$$A_f < 2Q^2 \quad (6)$$

$$A_f < 2(10.4)^2 = 216.32 \quad (7)$$

Taking,

$$A_f = 200 \quad (8)$$

Calculation of R_3 :

$$R_3 = \frac{Q}{\pi f_c C} \quad (9)$$

$$R_3 = \frac{10.4 \times 10^6}{3.14 \times 5200 \times 0.01} = 63.69 \text{ k}\Omega \quad (10)$$

$$R_1 = \frac{Q}{2\pi f_c C A_f}$$

$$R_1 = \frac{10.4 \times 10^6}{2 \times 3.14 \times 5200 \times 0.01 \times 200} = 159.23 \Omega$$

Practical Value Taken:

$$R_3 = (47 + 15) \text{ k}\Omega \quad (11)$$

Calculation of R_1 :

$$(12)$$

$$(13)$$

Practical Value Taken:

$$R_1 = (150 + 10) \Omega \quad (14)$$

Calculation of R_2 :

$$R_2 = \frac{Q}{2\pi f_c C (2Q^2 - A_f)} \quad (15)$$

$$R_2 = \frac{10.4 \times 10^6}{2 \times 3.14 \times 5200 \times 0.01 (2(10.4)^2 - 200)} = 1957.41 \Omega \quad (16)$$

Practical Value Taken:

$$R_2 = (1500 + 470) \Omega \quad (17)$$

0.6 Conclusion

The BPSK modulation and demodulation experiment successfully demonstrated the principle of binary phase shift keying, where a carrier wave's phase is altered to represent binary data. The modulation stage effectively shifted the phase by 180° based on the input message signal, while the demodulation circuit, using envelope detection and a comparator, accurately reconstructed the original signal. This experiment highlights BPSK's robustness in digital communication, offering efficient bandwidth utilization and noise immunity, making it a widely used modulation technique in modern wireless and satellite systems.

0.7 Precautions

1. Check the connections before switching on the kit.

2. Connections should be done properly.
3. Observation should be taken properly.